

Gym Membership and Fitness Tracking System

Project Team Information:

Team Name: Hex Vision

Team Members: Abdul Samee, Abdul Rafay, Amman Haroon

Team Logo:



1. Project Introduction

The Gym Membership and Fitness Tracking System is a web-based application designed to manage gym memberships, class bookings, trainer assignments, and personal fitness tracking for members. The system provides an intuitive, responsive interface for members, trainers, and admins, ensuring secure authentication, membership management, workout tracking, and administrative control.

2. Software Project Plan

2.1 Project Overview

The system provides gym members access to membership management, trainer booking, and fitness progress tracking while enabling trainers to manage assigned clients and administrators to oversee gym operations.

2.2 Project Objectives

- Allow members to sign up, view memberships, track fitness progress.
- Enable trainers to provide client programs.
- Provide admins tools to manage memberships, trainers, and classes.
- Ensure secure authentication and role-based access control.
- Deliver a clean, responsive user interface.

2.3 Project Scope

In Scope:

- Member registration, login, membership selection.
- Workout log tracking.
- Trainer assignment and management.
- Admin user and class management.
- Notifications and progress tracking.

Out of Scope:

- Payment gateway integration.

2.4 Project Deliverables

- Full stack web application (React.js frontend, Node.js backend)
- Complete source code repository
- Full documentation (user manual, developer guide)
- Testing reports
- Final project report (this document)

2.5 Project Organization and Roles

- Project Manager: Oversees project activities.
- Frontend Developer: Develops React interface.
- Backend Developer: Implements Node.js API and database operations.
- Tester/QA: Develops and runs test cases.
- All: Participate in documentation and project reviews.

2.6 Risk Management

- Technical Risks: React learning curve.
- Schedule Risks: Delays in feature implementation.
- Mitigation: Early prototyping, buffer periods.

2.7 Communication Plan

- Weekly meetings.
- Daily updates through group chats.
- Trello board for task management.

2.8 Quality Assurance Plan

- Code reviews
- Automated and manual testing
- Minimum 70% code coverage target

2.9 Tools and Technologies

- Frontend: React.js
- Backend: Node.js, Express.js
- Database: MySQL
- Testing: Jest
- Project Management: Trello, GitHub

2.10 Success Criteria

- All functionalities operational.
- Smooth, secure, responsive experience.
- User acceptance testing passed.
- Complete documentation.

3. Work Breakdown Structure (WBS)

Level 1: Project

Gym Membership and Fitness Tracking System

Level 2: Deliverables

- Web Application
- Database System
- Documentation

Level 3: Components

- User Management Module
- Membership Management Module

- Trainer Management Module
- Booking System Module
- Fitness Tracker Module

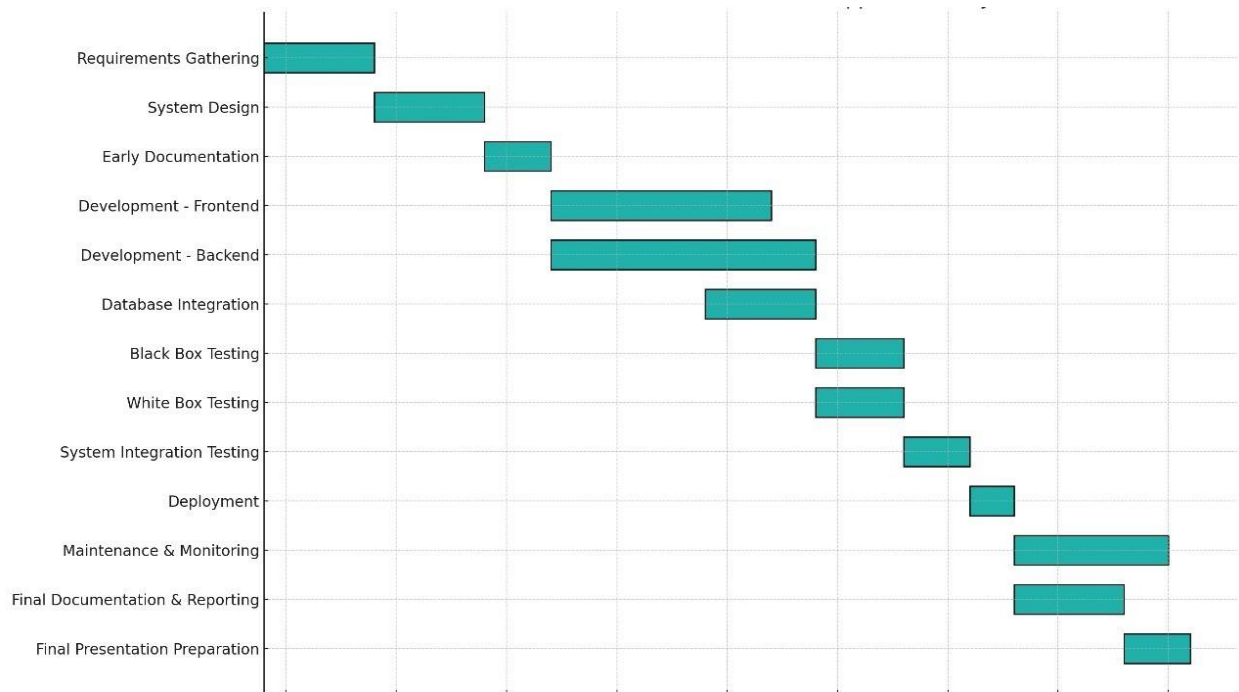
Level 4: Activities

- Design Frontend Screens
- Develop Backend
- Integrate Frontend and Backend
- Test Module Functionality
- Write User Documentation

Level 5: Tasks

- Create Login Form Validation
- Implement Registration
- Connect MySQL Database to Backend
- Build Trainer Profile Page
- Implement Session Booking Logic
- Setup Fitness Logging UI
- Test Membership Renewal Logic

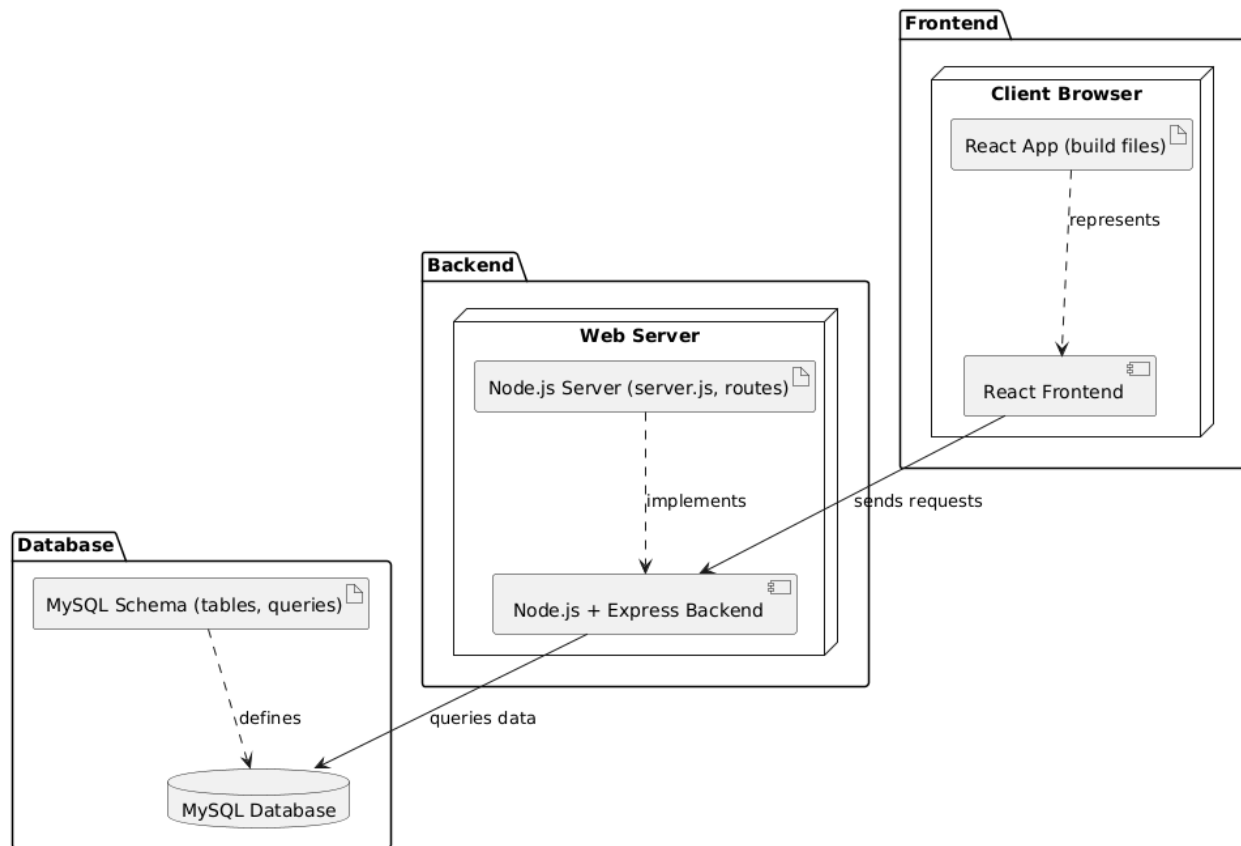
3.1. Gantt Chart



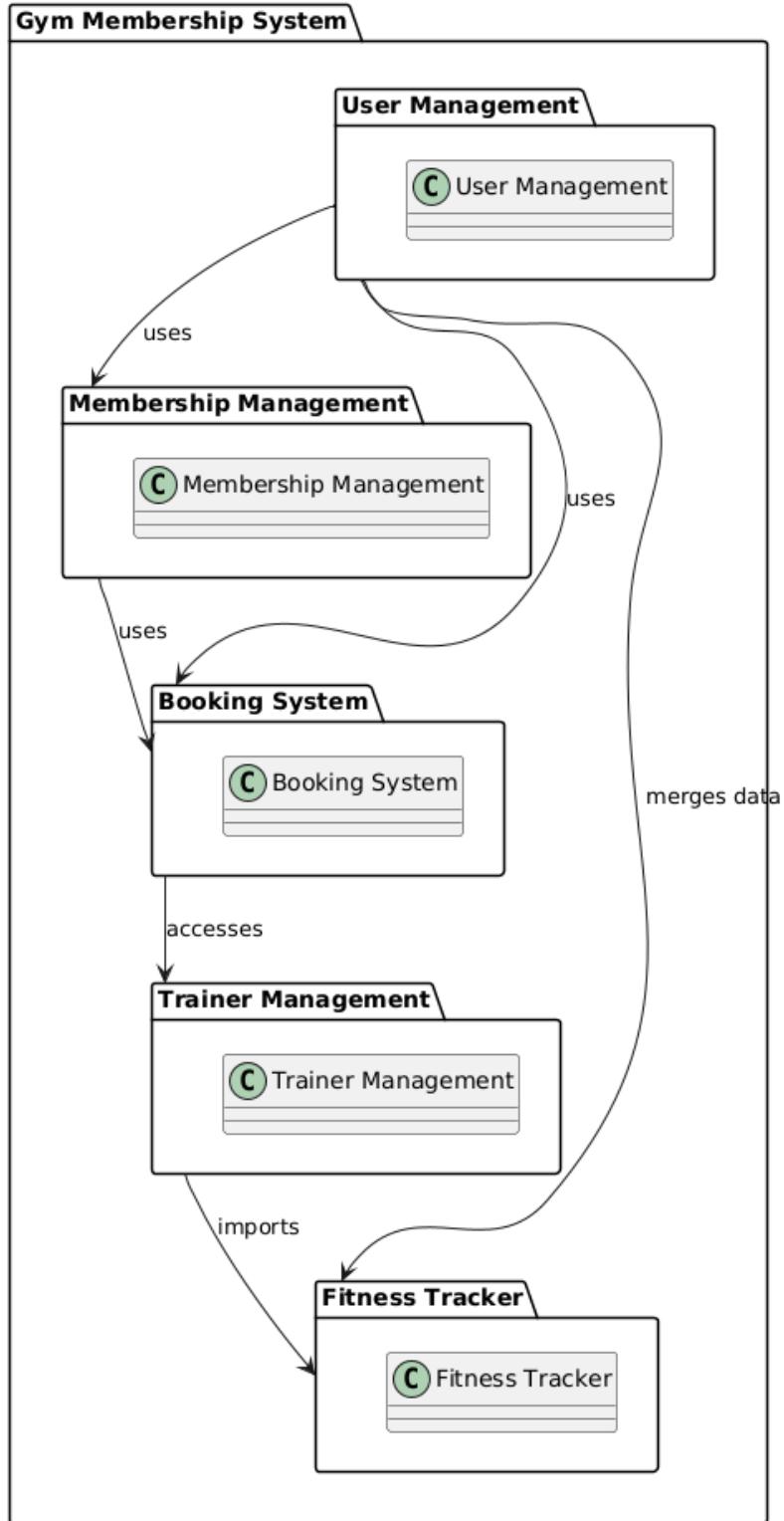
4. Architecture Styles

- **Layered Architecture:** Separation between frontend (React.js), backend (Node.js/Express), and database (MySQL).
- **Client-Server Architecture:** Clients (browsers) interact with a centralized server for all operations.

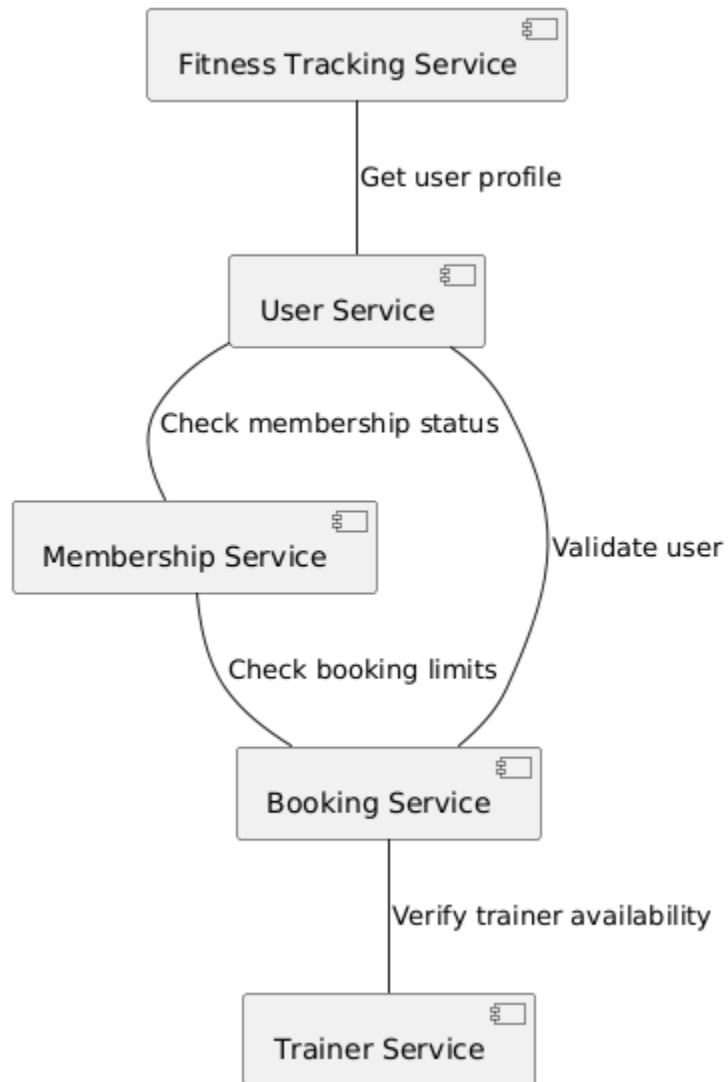
4.1. Deployment Diagram



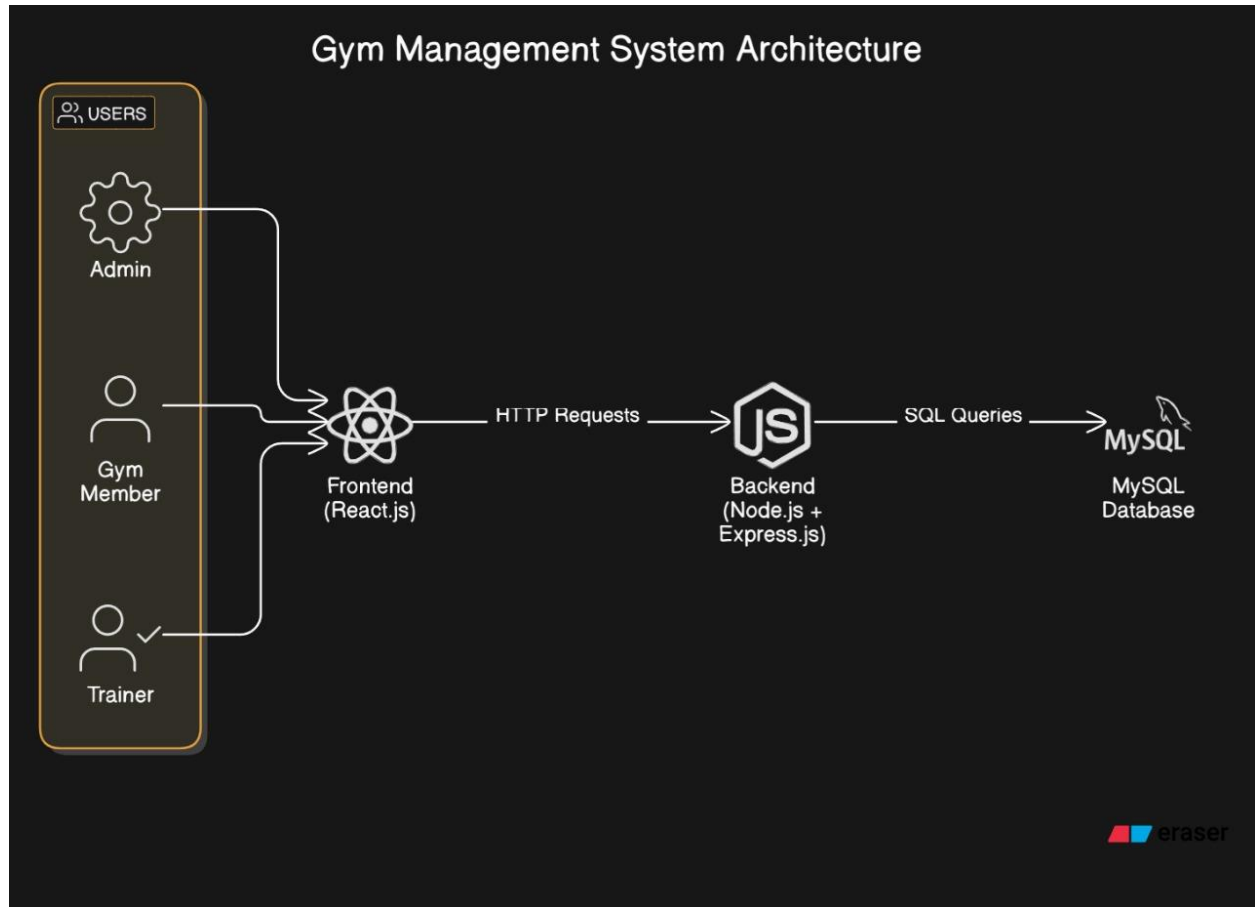
4.2. Package Diagram



4.3. Component Diagram



4.4. Architecture Diagram



5. System Functional Requirements

The system shall provide the following functionalities for each stakeholder:

- **Gym Member:** -

1. The system shall allow users to register and log into their account using secure authentication.
2. The system shall allow users to view and manage their membership details, including start date, end date, and renewal options.
3. The system shall allow users to book workout sessions with trainers based on their availability and expertise.
4. The system shall allow users to log their workout progress, including exercise type, duration, and calories burned.
5. The system shall provide users with a dashboard to view their fitness progress over time.

- **Trainer:** -

1. The system shall allow trainers to register and create a profile including their expertise, availability, and contact information.
2. The system shall allow trainers to view and manage their assigned clients.
3. The system shall allow trainers to update fitness plans and workout schedules for clients.
4. The system shall allow trainers to view their schedule of booked workout sessions.
5. The system shall allow trainers to provide feedback to users based on their workout progress.

- **Admin:** -

1. The system shall allow admins to manage user accounts by activating or deactivating them.
2. The system shall allow admins to update membership plans, including pricing and features.
3. The system shall allow admins to oversee the booking system and resolve conflicts if necessary.
4. The system shall allow admins to generate system reports and analytics.
5. The system shall allow admins to monitor the overall system performance and manage backup and recovery processes.

6. System Non-Functional Requirements

6.1 Product Requirements: -

1. The system shall be accessible to users, trainers, and admins via a web-based interface built with React.js.
2. The system shall maintain availability 100% of the time.
3. The system shall respond to user actions within 2 seconds under normal conditions.

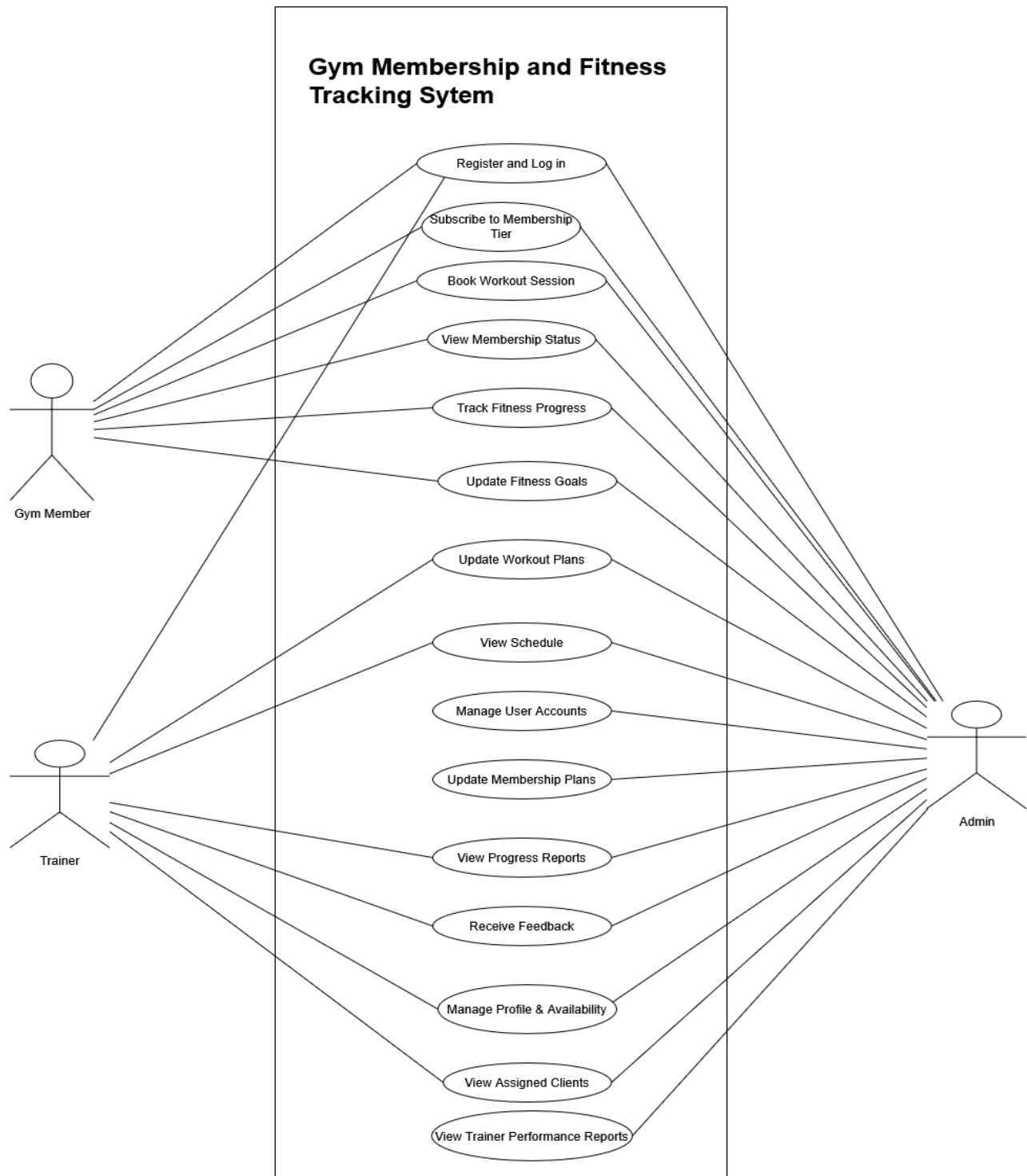
6.2 Organizational Requirements: -

1. The system shall be developed using React.js (Frontend), Node.js with Express.js (Backend), and MySQL (Database).
4. Documentation for the system shall include installation, deployment, and maintenance guides.

6.3 External Requirements: -

1. The system shall comply with data privacy standards to protect user information.

7. Use Case Diagram: -



8. User Stories: -

1. As a user, I want to register and log in to my account so I can access my gym membership and fitness details.

- **Role:** User (Gym Member)
- **Goal:** To register and log in to their account.
- **Reason:** So, they can access their gym membership and fitness details.
- **Pre-condition:** User is not registered or is not logged in.
- **Post-condition:** User is successfully registered and logged in.

2. As a user, I want to subscribe to a membership tier so I can use its respective perks.

- **Role:** User (Gym Member)
- **Goal:** To subscribe to a membership tier.
- **Reason:** To gain access to its respective perks.
- **Pre-condition:** User is logged in but has no active membership.
- **Post-condition:** User is subscribed to a chosen membership plan.

3. As a user, I want to view my membership status so I can check my expiration date and renewal details.

- **Role:** User (Gym Member)
- **Goal:** To view their membership status.
- **Reason:** To check expiration date and renewal details.
- **Pre-condition:** User is logged in and has an active membership.
- **Post-condition:** User views membership status successfully.

4. As a trainer, I want my profile to include my expertise and availability so users can choose me based on their needs.

- **Role:** Trainer
- **Goal:** To include expertise and availability in their profile.
- **Reason:** So, users can choose trainers based on their needs.
- **Pre-condition:** Trainer is not registered or has not created a profile.
- **Post-condition:** Trainer profile is successfully created and accessible.

5. As a trainer, I want to view a list of my assigned clients so I can manage their fitness plans.

- **Role:** Trainer
- **Goal:** To view a list of assigned clients.
- **Reason:** To manage clients' fitness plans effectively.
- **Pre-condition:** Trainer is logged in and has clients assigned.
- **Post-condition:** List of clients is successfully displayed

6. As a user, I want to securely pay for my gym membership online using my preferred payment method so that I can activate or renew my membership without visiting the gym physically.

- **Role:** User (Gym Member)
- **Goal:** To make an online payment for membership.
- **Reason:** To activate or renew their membership without physically visiting the gym.
- **Pre-condition:** User is logged in and has selected a membership plan.
- **Post-condition:** Payment is processed successfully, and membership is activated or renewed.

7. As a user, I want to log my current weight, body fat percentage, and daily water intake so that I can monitor my fitness progress over time using visual graphs.

- **Role:** User (Gym Member)
- **Goal:** To log fitness metrics such as weight, body fat, and water intake.
- **Reason:** To track and visualize progress over time.
- **Pre-condition:** User is logged in and has an active membership.
- **Post-condition:** Fitness data is successfully logged and displayed through visual graphs.

8. As a user, I want to book a workout program with my trainer so I can work on my fitness goal.

- **Role:** User (Gym Member)
- **Goal:** To book a workout program with a trainer.
- **Reason:** To work on their fitness goals.
- **Pre-condition:**
 1. User is logged in and has an active membership.
 2. Available trainers and slots are listed.
- **Post-condition:** Workout program is successfully booked.

9. As a user, I want to set specific fitness goals such as increasing muscle mass or reducing body fat so that I can stay motivated and track my progress toward those goals with completion percentages.

- **Role:** User (Gym Member)
- **Goal:** To set and monitor specific fitness goals.
- **Reason:** To stay motivated and track progress toward achieving goals.
- **Pre-condition:** User is logged in and has an active membership.
- **Post-condition:** Goals are set, and progress is tracked with completion metrics.

10. As a trainer, I want to receive feedback from clients so I can improve my training methods.

- **Role:** Trainer
- **Goal:** To receive feedback from clients.
- **Reason:** To improve training methods.
- **Pre-condition:** Trainer is logged in and has clients assigned.
- **Post-condition:** Feedback from clients is successfully received.

11. As a user, I want to track my fitness progress so that I can monitor improvements.

- **Role:** User (Gym Member)
- **Goal:** To track fitness progress.
- **Reason:** To monitor improvements.
- **Pre-condition:** User is logged in and has an active membership.
- **Post-condition:** Fitness progress is successfully recorded and stored.

12. As a user, I want to log detailed workout sessions including exercises, sets, reps, and notes so that I can track improvements and review my past training history.

- **Role:** User (Gym Member)
- **Goal:** To log workout session details.
- **Reason:** To track improvements and maintain a training history.
- **Pre-condition:** User is logged in and has an active membership.
- **Post-condition:** Workout session is logged and stored for future reference.

13. As an admin, I want to manage user accounts so I can control system access.

- **Role:** Admin

- **Goal:** To manage user accounts.
- **Reason:** To control system access effectively.
- **Pre-condition:**
 1. Admin is logged in with proper permissions.
 2. User accounts exist in the system.
- **Post-condition:** User account status is updated (activated or deactivated).

14. As an admin, I want to add or remove gym membership plans and pricing so users see the latest options.

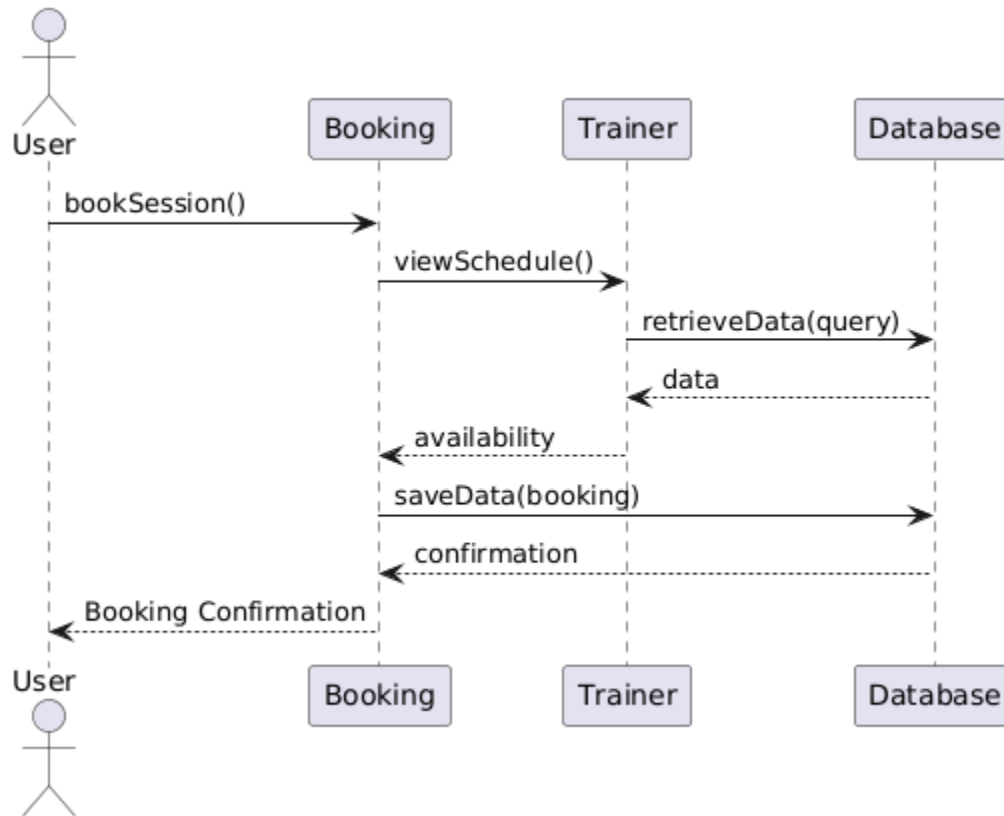
- **Role:** Admin
- **Goal:** To add or remove gym membership plans and pricing.
- **Reason:** So, users see the latest options.
- **Pre-condition:** Admin is logged in and has permission to update plans.
- **Post-condition:** Membership plans and pricing are updated.

15. As an admin, I want to view trainer performance feedback to ensure quality service.

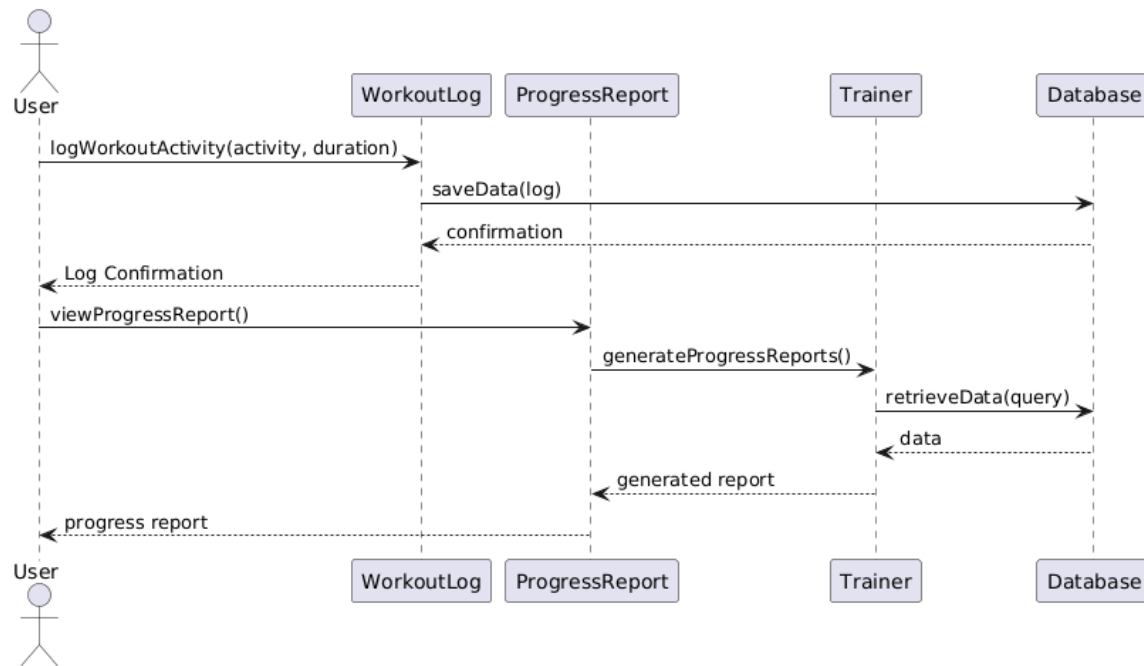
- **Role:** Admin
- **Goal:** To view trainer performance feedback.
- **Reason:** To ensure quality service.
- **Pre-condition:**
 1. Admin is logged in with proper permissions.
 2. Trainer performance data exists in the system.
- **Post-condition:** Trainer performance reports are successfully displayed.

9. Sequence Diagrams: -

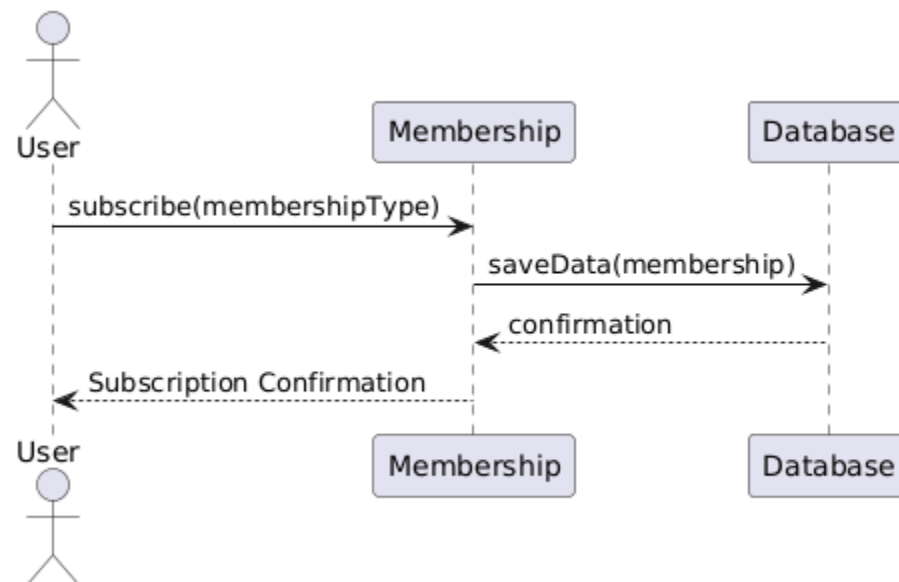
1. Book Workout Session



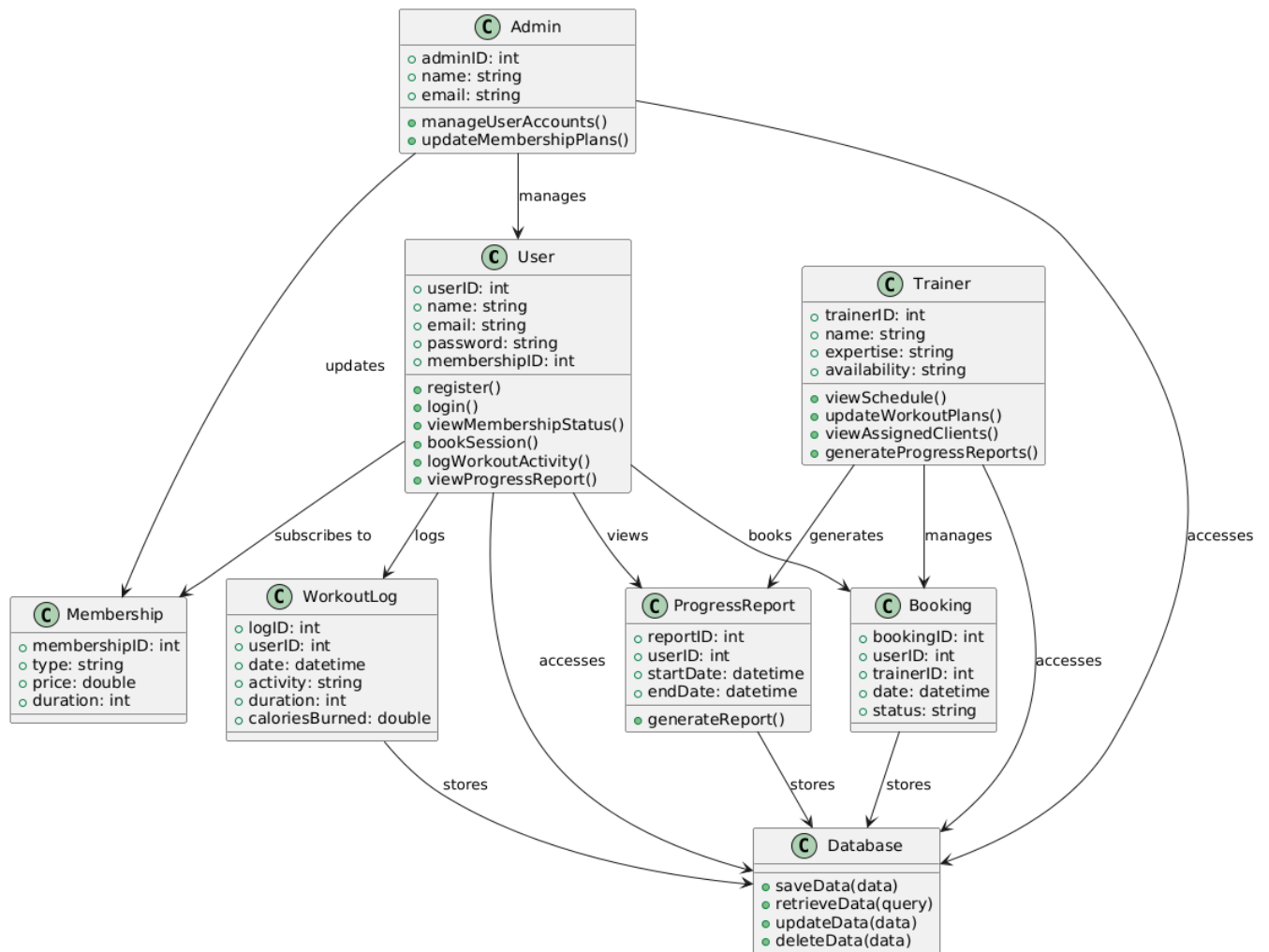
2. Fitness Tracking



3. Membership Subscription

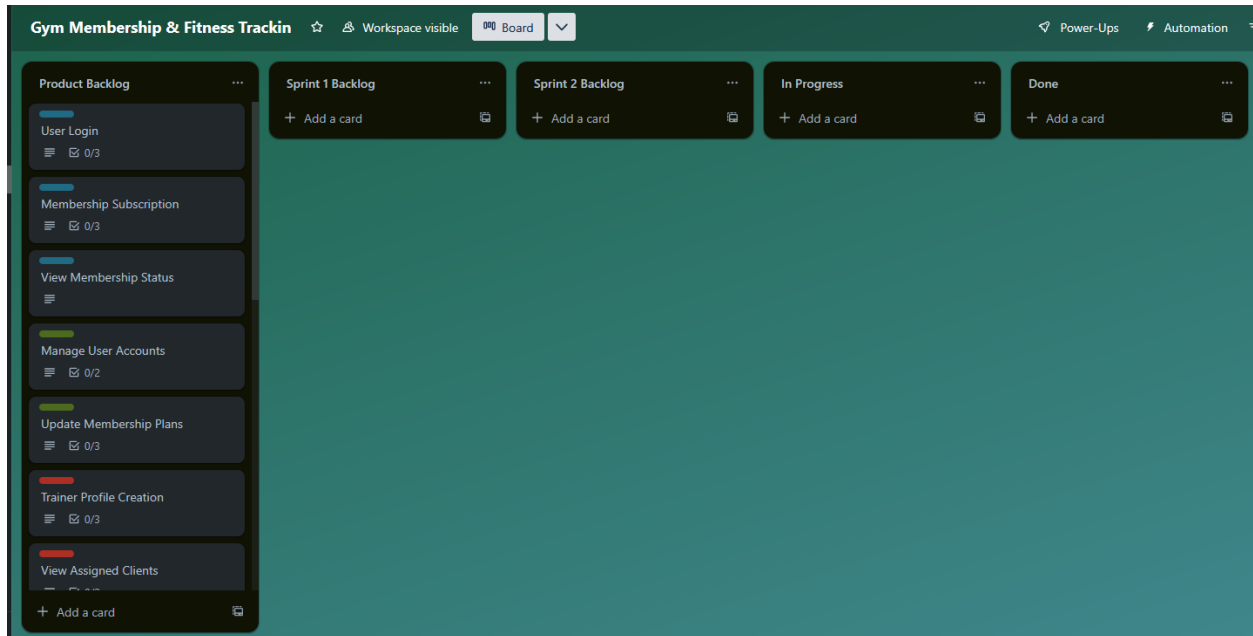


10. Class Diagram: -

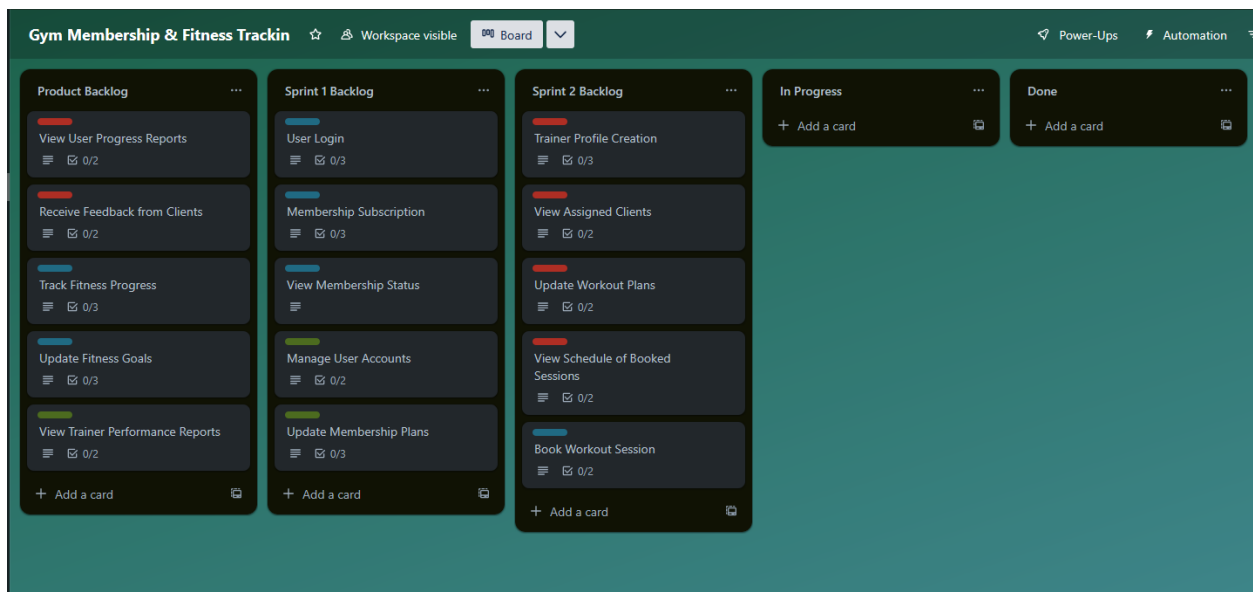


11.Trello Board: -

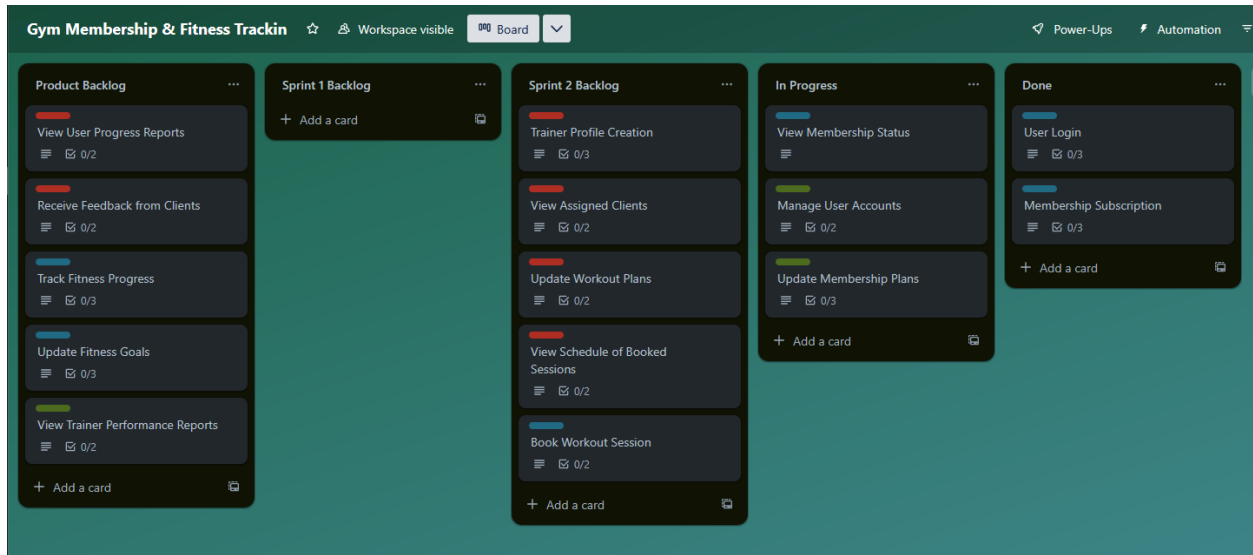
Product Backlog



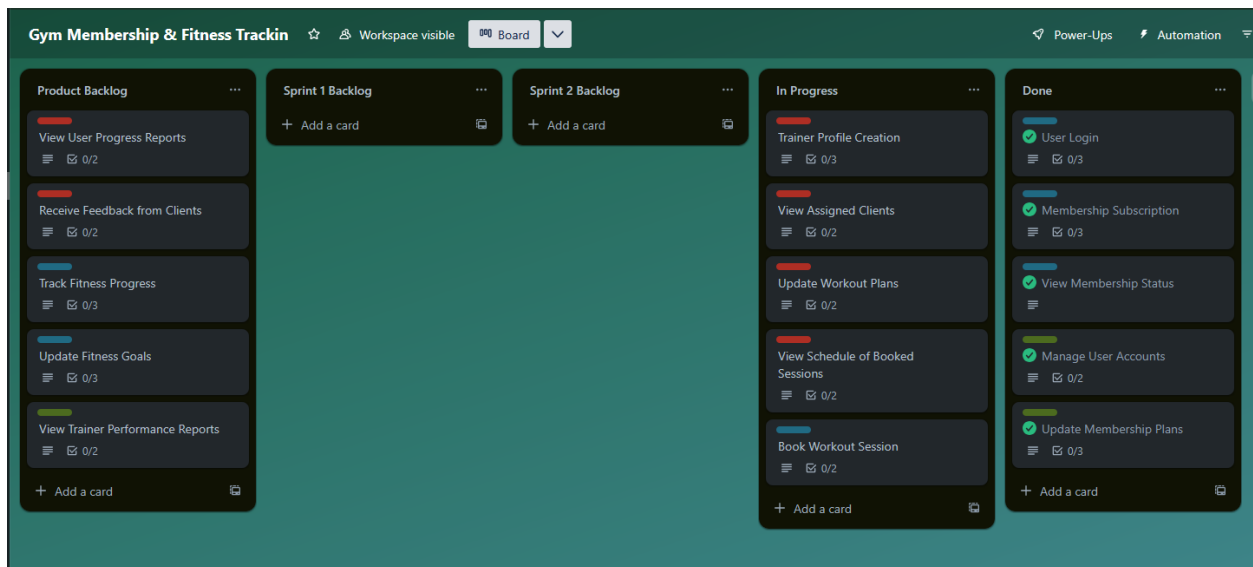
Sprint 1 and Sprint 2 Backlog



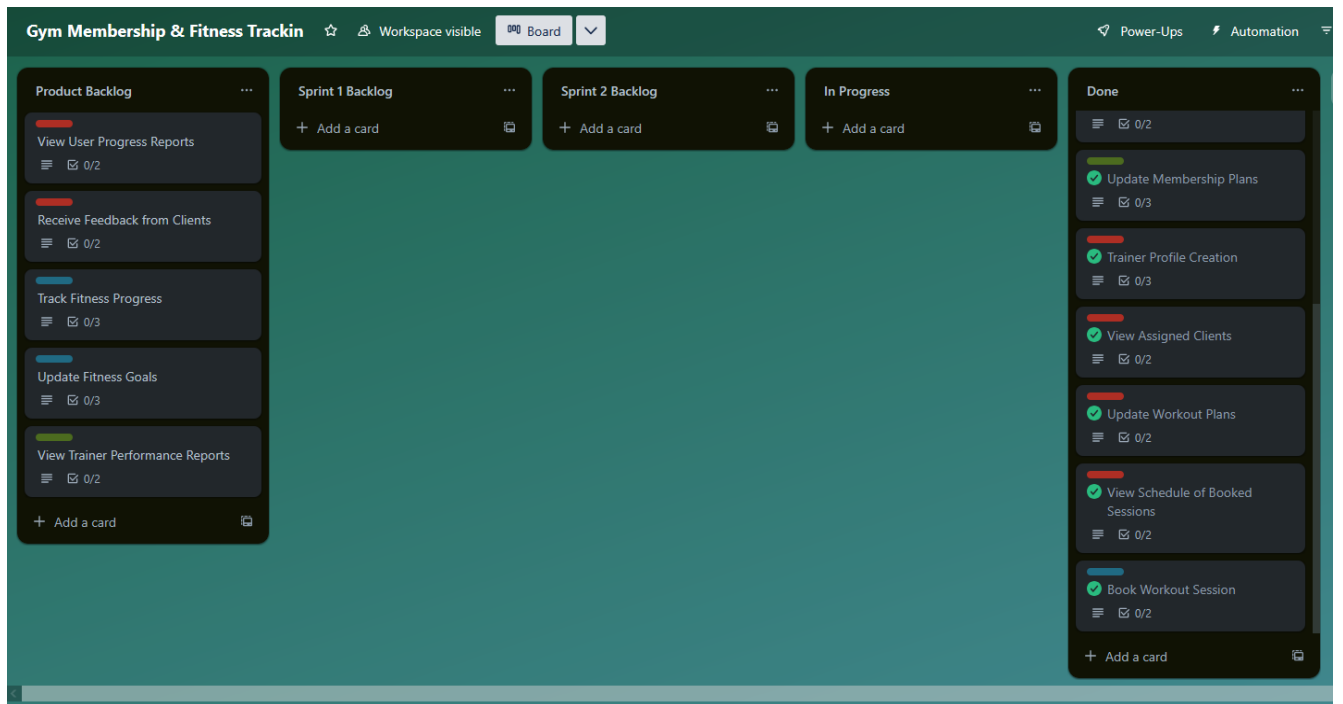
Sprint 1 In Progress



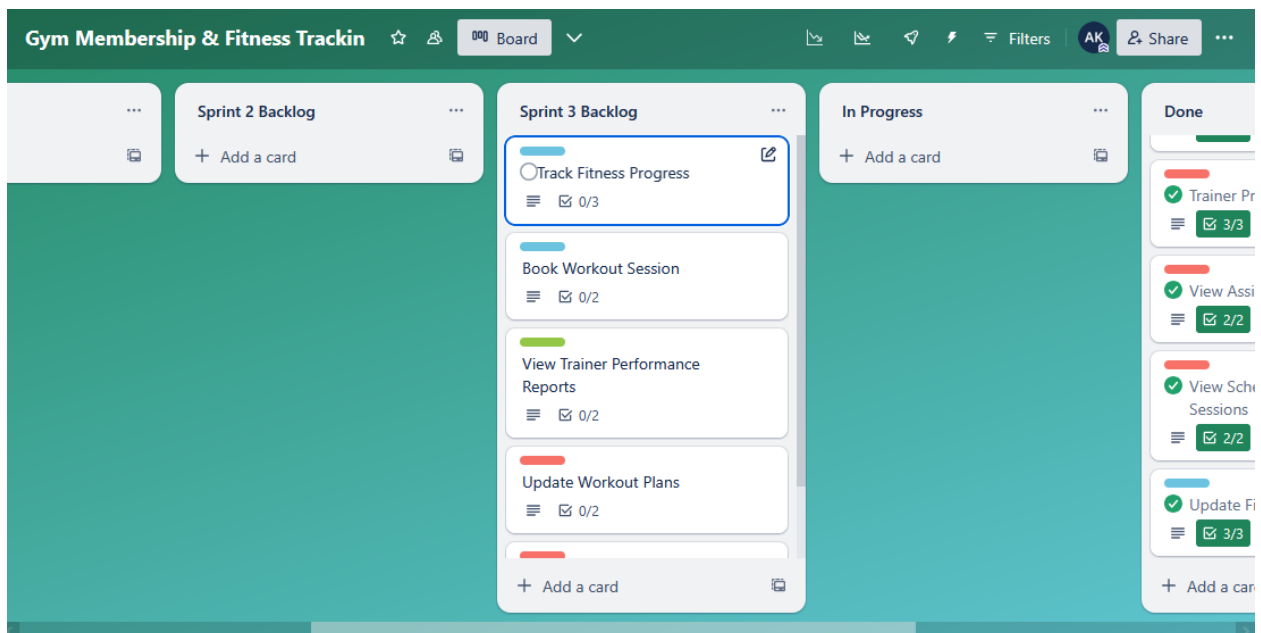
Sprint 2 In Progress



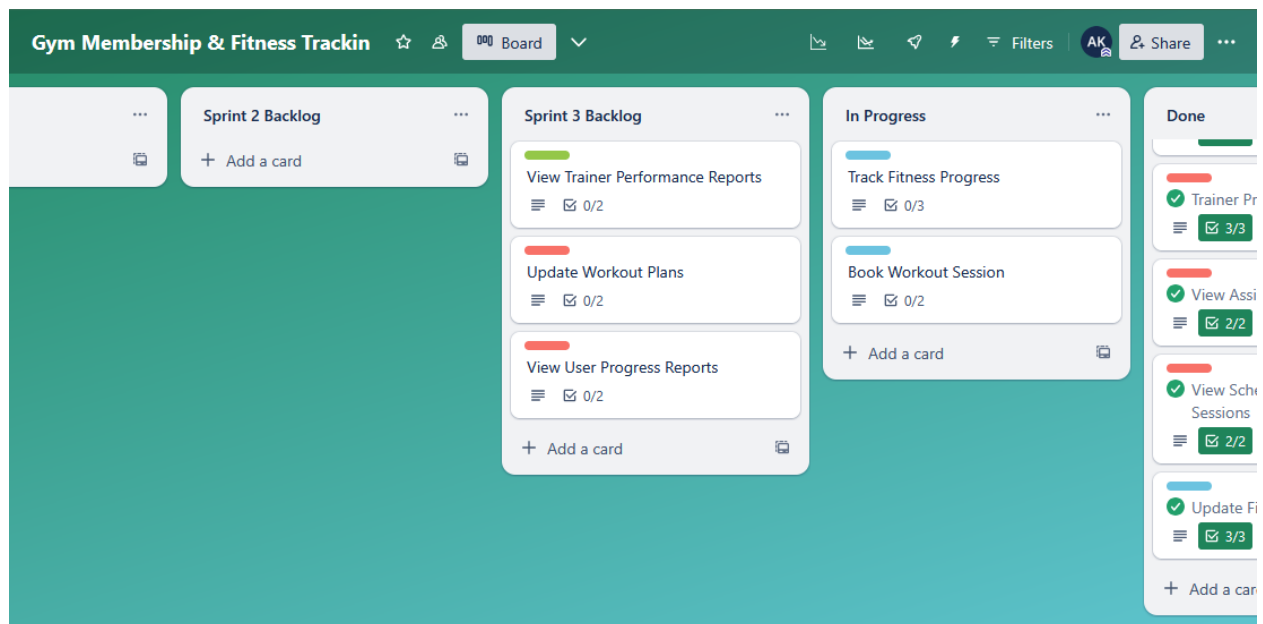
Sprint 1 And Sprint 2 Done



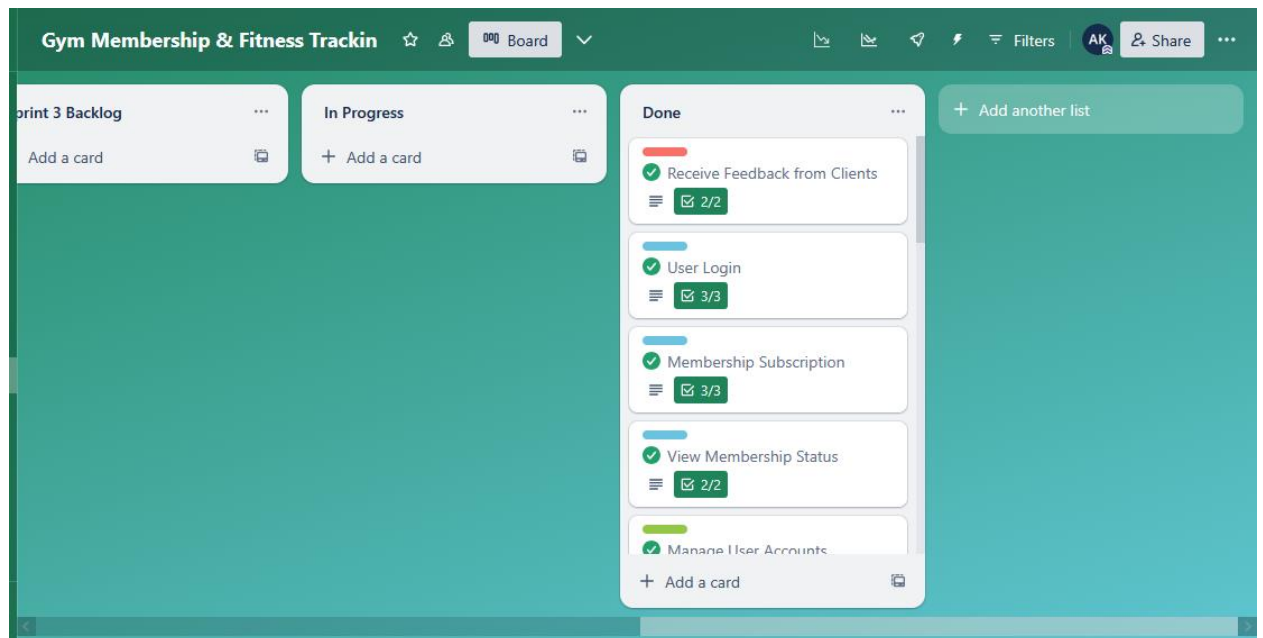
Sprint 3 Backlog



Sprint 3 In Progress



Project Done



12. Project Plan

The project is planned and executed in three main phases:

1. Requirements and Design:

- Gathered requirements.
- Defined user stories and acceptance criteria for core features.
- Designed system architecture and database schema, including entities for users, memberships, fitness tracking, and trainer management.
- Created all necessary UML diagrams (Use Case Diagram, Sequence Diagrams, Class Diagram).

2. Development and Testing:

- Developed backend using Node.js with Express.js and MySQL, implementing role management, and CRUD operations for gym members, trainers, and admins.
- Developed frontend using React.js ensuring responsive design and user-friendly dashboards.
- Integrated frontend and backend.
- Implemented core features: Membership registration, trainer management, booking system, and fitness progress tracking.
- Performed black box and white box testing for UI components, line coverage and integration.
- Achieved 70%+ code coverage with unit tests and integration tests for critical modules.

3. Deployment and Review:

- Performed final feature enhancements.
- Prepared documentation, including final report, for project submission.

13.Design

1. Register and Log In to Account

- **UI Design:**

- Registration form (name, email, password, confirm password)

- Login form (email, password)
- **System Design:**
 - Authentication module (handles login sessions)
 - User Manager (handles creation, encryption, login verification)
- **Database Tables:**
 - users: id, name, email, password_hash, role, created_at

2. Subscribe to a Membership Tier

- **UI Design:**
 - Membership plans listing
 - Subscribe/Choose button for each tier
- **System Design:**
 - Membership Manager (handles tier logic and association with user)
- **Database Tables:**
 - memberships: id, name, features, price, duration_days
 - user_memberships: id, user_id, membership_id, start_date, end_date, status

3. View Membership Status

- **UI Design:**
 - Dashboard widget showing plan name, start date, end date, renewal status
- **System Design:**
 - Membership Display Logic (calculates days left, renewal alerts)

4. Admin – Manage User Accounts

- **UI Design:**

- Admin dashboard with user list
 - Action buttons (Activate, Deactivate)
 - **System Design:**
 - Admin module (user status management logic)
 - **Database Updates:**
 - users: add column is_active (boolean)
-

5. Admin – Add/Remove Membership Plans

- **UI Design:**
 - Form to add new membership plans
 - List of plans with delete/edit actions
 - **System Design:**
 - Plan Manager (validates and updates membership catalog)
-

6. Trainer – Profile with Expertise and Availability

- **UI Design:**
 - Trainer profile form: expertise (text), availability (calendar/times)
 - Public profile preview for users
 - **System Design:**
 - TrainerProfile module
 - **Database Tables:**
 - trainer_profiles: id, user_id, expertise, availability_json, created_at
-

7. Trainer – View Assigned Clients

- **UI Design:**

- Trainer dashboard showing assigned clients (name, goal, start date)
 - **System Design:**
 - Trainer-Client Assignment Logic
 - **Database Tables:**
 - trainer_clients: id, trainer_id, user_id, assigned_at
-

8. User – Book Workout Program

- **UI Design:**
 - Booking page showing available trainers and time slots
 - Book button with confirmation modal
- **System Design:**
 - Booking Manager (ensures slot availability and assigns trainer)
- **Database Tables:**
 - workout_bookings: id, user_id, trainer_id, datetime, status

9. User – Secure Online Payment for Membership

- **UI Design:**
 - Payment form with method selection (credit card, etc.)
 - Payment summary and confirmation
 - **System Design:**
 - Payment Handler (updates membership after confirmation)
 - **Database Tables:**
 - payments: id, user_id, membership_id, amount, method, status, date_paid
-

10. User – Log Weight, Body Fat, Water Intake

- **UI Design:**

- Fitness metrics entry form
 - Graphs to visualize trends over time
 - **System Design:**
 - FitnessLog Manager (calculates and stores trends)
 - **Database Tables:**
 - fitness_logs: id, user_id, weight, body_fat, water_intake, log_date
-

11. User – Set and Monitor Fitness Goals

- **UI Design:**
 - Goal setup form (goal type, target, timeframe)
 - Progress bar showing % achieved
 - **System Design:**
 - Goal Tracker (calculates progress from logs)
 - **Database Tables:**
 - fitness_goals: id, user_id, goal_type, target_value, current_value, start_date, end_date
-

12. Trainer – Receive Client Feedback

- **UI Design:**
 - Feedback submission form (rating, comment)
 - Trainer view of received feedback
 - **System Design:**
 - Feedback Manager
 - **Database Tables:**
 - trainer_feedback: id, trainer_id, user_id, rating, comment, submitted_at
-

13. User – Track Fitness Progress (Graph View)

- **UI Design:**
 - Fitness dashboard with:
 - Weight trend
 - Body fat trend
 - Water intake trend
 - **System Design:**
 - Graph Engine (processes fitness_logs to show trends)
-

14. User – Maintain Workout Logs

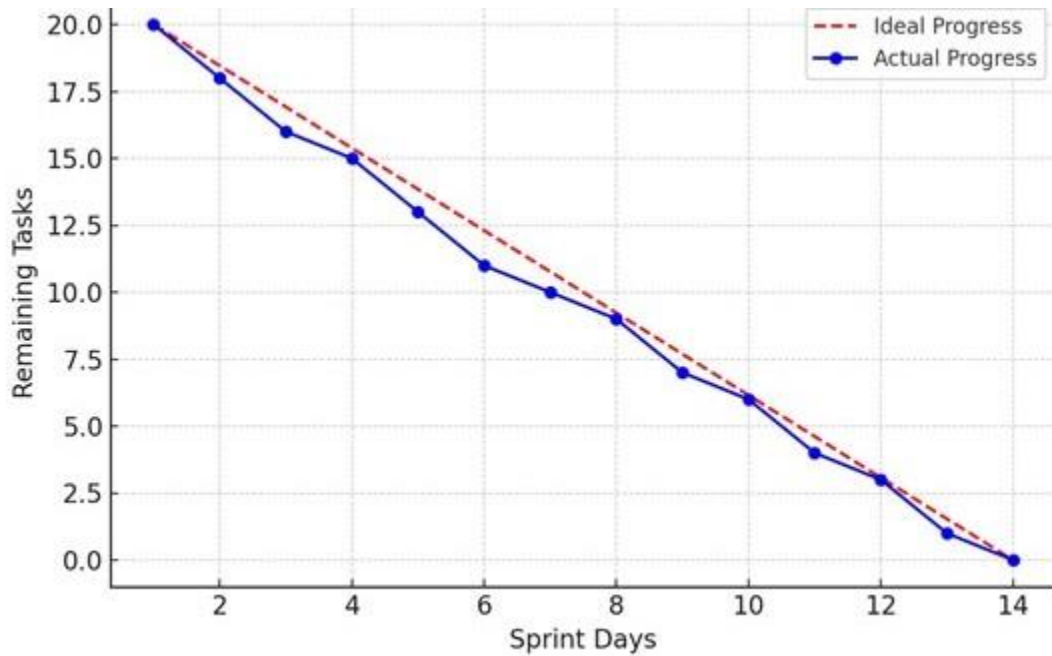
- **UI Design:**
 - Workout log entry form (exercise, sets, reps, weight, notes)
 - Table or list of past workouts
 - **System Design:**
 - WorkoutHistory Manager
 - **Database Tables:**
 - workout_logs: id, user_id, exercise, sets, reps, weight, notes, log_date
-

15. Admin – View Trainer Feedback Reports

- **UI Design:**
 - Admin dashboard panel for trainer feedback summary
 - Filter by trainer, rating, date range
- **System Design:**
 - Report Generator (aggregates trainer_feedback)
- **Database Query:**

- Aggregation on trainer_feedback for performance insights

14. Burn Down Chart



15. Work Division

Abdul Rafay:

Backend Development (Node.js, MySQL), Project Management, Frontend Development (React.js)

Amman Haroon

Backend Development (Node.js, MySQL), Frontend Development (React.js), Integration

Abdul Samee

Frontend Development (React.js), Documentation, Testing

16. Lesson Learnt

- Early prototyping and consistent communication were key to progress.
- Gained hands-on experience with React, Node.js, and MySQL integration.
- Improved collaboration using Trello for task management and GitHub for version control.